

Week 3: Seller & Inventory Management System - Documentation

1. Seller & Inventory Table Design

- **Seller Table:** Seller_ID (PK), Seller_Name (NOT NULL), Email (UNIQUE), Phone, Address.
- **Inventory Table:** Inventory_ID (PK), Product_ID (FK), Seller_ID (FK), Stock_Quantity (CHECK >= 0), Stock_Status, Last_Updated.
- **Product Table:** Product_ID (PK), Product_Name, Category_ID (FK), Price (CHECK > 0).

2. Updated ER Diagram Representation

Seller (1) -----< (M) Inventory (M) >----- (1) Product ----- (1) Category

- Seller relates to Inventory via Seller_ID.
- Product relates to Inventory via Product_ID and Category via Category_ID.

3. Sample Seller & Inventory Data Summary

Inv ID	Product	Seller	Stock	Status
1	Smartphone X	TechCorp Global	50	In Stock
2	Wireless Earbuds	TechCorp Global	120	In Stock
3	Gaming Laptop	TechCorp Global	5	Low Stock
4	Cotton T-Shirt	FashionHub Retail	200	In Stock
5	Denim Jeans	FashionHub Retail	80	In Stock
6	Blender Pro	HomeEssentials Inc	0	Out of Stock
7	Coffee Maker	HomeEssentials Inc	30	In Stock
8	Smart TV 55 in	TechCorp Global	15	In Stock

4. Inventory Analysis Report

- **Stock Health:** Majority of items are in stock, ensuring high fulfillment rate.
- **Supplier Distribution:** TechCorp Global supplies high-value electronics, whereas FashionHub and HomeEssentials manage apparel and kitchenware.
- **Low Stock & Replenishment:** Items like Gaming Laptop required immediate stock updates.
- **Average Inventory:** The average stock quantity per inventory item stands at approximately 62.5 units.

5. Query Output & Verification Screenshots

All SQL statements for table creation, multi-table joins, constraints enforcement, seller CRUD, inventory tracking, and aggregate reporting have been successfully tested and executed in the database.